

Editorial

ICSSP 2011 Special issue: processes for tomorrow's systems and software engineering: an evolving dynamic domain

Welcome to the special issue containing the best papers from the 2011 International Conference of Systems and Software Process. This 2011 conference is both a *continuation* and an *evolution* of the successful ICSP and ProSim conference series. It's a *continuation* in the sense of being the leading international academic conference devoted to Process related research. It's also a *continuation* in the sense that the conference seeks to provide an outlet for that special blend of academic and applied work which has a high degree of impact. ICSSP is an *evolution* in the sense that this year and for the years to come, this conference will focus on both Systems and Software Process. With this intentional incorporation of Systems, there will be a broadening to embrace *process issues* associated with all of the disciplines that are part of the Systems Engineering enterprise. With this larger vision, we begin taking steps to integrate these important disciplines and make a more meaningful contribution and impact to the academic and practitioner communities at large.

The papers included in this issue were ranked highest by reviewers among all the papers submitted to the ICSSP 2011 conference. In all, 102 papers were submitted to the conference. From that 28 papers were accepted for presentation and of those 6 were selected for this special issue. While many of the papers did focus on Software Process specifically, we sought to select papers for this special issue that have meaningful insights for the broader Systems Engineering Community as well.

We begin the special issue with "Integrating Inspection and Test Processes based on Context-Specific Assumptions", by Elberzhager, Münch, Rombach and Freimut. Inspections and testing are two of the most prevalent quality assurance activities. This paper describes the improved efficiency and effectiveness possible when these two processes are systematically integrated and discusses how to accomplish this. The integration enables context specific knowledge about the product to be utilized. The end result is improved quality, reduced schedule and less effort. Integrating the inspection and testing processes at the Systems Engineering level has the potential to produce significant increases in both efficiency and effectiveness.

The second paper is "MDE-Based Process Tailoring Strategy" by Alegría, Bastarrica, Quispe, and Ochoa. This paper describes a systematic approach to tailoring organizational processes to project particulars. It is well known that tailoring is important to project success. However, it is challenging because of the number and variety of details that are specific to the project. This paper develops a framework and shows examples for how this tailoring can be done for the requirements engineering process at a medium sized Chilean company.

The third paper is titled, "Agile Coaching for Global Software Development" by Paasivaara and Lassenius. Coaching project teams is a critical part of successful deployment of agile practices within an organization. This paper deals with the coaching of globally distributed agile development projects and focuses on how to build and sustain a successful team of coaches. Specifically this paper describes the main challenges of building a team of coaches as well as the agile practices used by the coaches as part of their coaching activities. The paper is based on semi-structured interviews of coaching team members and project personnel from multiple coached projects. The paper concludes with a set of lessons learned for coaching globally distributed agile projects.

The fourth paper is titled, "Embedded Software Product Lines: Domain and Application Engineering Model Based Analysis Processes" by Belategi, Sagardui, Etxeberria, and Azanza. This paper focuses on embedded systems development by presenting a process based on UML and MARTE to analyze embedded model-driven product-lines. The model that is presented defines tasks, roles, work products and existing tools used to support the process as well as how the

product-line portion of the process integrates into the more general development process. The paper also provides a classification of the tools and the traceability among them which allows systems engineering teams to choose the best tools to support the process at large.

The fifth paper is titled, “AVISPA: A Tool for Analyzing Software Process Models” by Alegría, Bastarrica, and Bergel. This paper describes a graphical tool called AVISPA that can be used to analyze the quality of SPEM 2.0 software process models. Specifically, the AVISPA tool identifies error patterns and blueprints found in the development process. Analysis of this kind is very important to improve the integrity, correctness, usability, maintainability, and performance of the process. This paper describes both the inner workings of the tool and how it can be extended to include additional patterns and blueprints increasing the scope and capabilities of the tool. In addition, the paper presents an industrial case study of applying the tool where process engineers were supported in analyzing the quality of their processes.

The sixth paper is titled, “An Empirical Study of Process Knowledge: Coherence as a Static Process Property” by Crabtree, Norcio, and Seaman. It has long been known that the *as-practiced* process often differs from the *as-documented* process or the *as-understood* process by process actors. Moreover, this presents a big problem with process compliance. With the help of a controlled experiment using a basic between-subject design, this paper studies the effects of the presence or absence of a conceptual model on the perception of errors between description and performance of a process. The results indicate that the presence of a conceptual model describing a process under consideration results in a higher perception of error.

We believe these papers present a good sampling of the work presented at the first ICSSP conference and that the work presented in these papers is relevant to both the Software and Systems Engineering communities. Further information about the ICSSP conference series as well as links to conference papers can be found at <http://www.icssp-conferences.org>.

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